

Empowering Ageing with Ingenuity: A Scoping Review on the Methods and Technologies Used into Elderly Health Monitoring

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Abstract. With the rise in global life expectancy, ensuring healthier aging experiences for the older population becomes paramount. This scoping review delves into the technologies employed in the remote health monitoring of the elderly over the past 15 years. Exploring the concept of "Healthy Ageing" as proposed by the World Health Organization, this paper attempts to highlight the techniques and technologies used in health monitoring of the elderly population. The integration of wearable sensors in health monitoring presents promising avenues for enhancing healthcare delivery to older adults. However, challenges such as limited digital literacy and privacy concerns persist, necessitating innovative solutions for unobtrusive monitoring. This paper discusses the potential of passive and ambient sensors to address these challenges, offering insights into enhancing the well-being of the older population while preserving their independence and privacy.

Keywords. Remote health monitoring, Older Adults, Healthy Ageing, Sensors, Unobtrusiveness

1. Introduction

Over the last century, advancements in healthcare and living conditions have significantly increased life expectancy worldwide. The global elderly population is expanding rapidly, with individuals aged 65 and above expected to increase from 10% in 2022 to 16% by 2050 [1]. This notable rise in life expectancy poses significant health challenges and is anticipated to result in considerable socio-economic burdens. In 2015, the WHO introduced a groundbreaking conceptual model for "Healthy Ageing," emphasizing functional ability over the mere presence or absence of diseases [2]. The UN along with the WHO introduced the decade of Healthy Ageing (2021-2030), with one of the key focus areas being providing integrated primary healthcare services and access to long-term care to the elderly population [1].

The landscape of health monitoring for older adults has undergone a profound transformation with the integration of wearable sensors. These devices, capable of real-

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time monitoring of vital signs, have attracted considerable attention for their potential to track various aspects of an individual's health and behavior [3]. Several academic and scientific projects have explored diverse implementations of wearable systems for healthcare monitoring, including smart textiles, implantable devices, epidermal devices, and other wearable and portable gadgets [4]. For instance, a smart wearable, facilitated with a wireless sensor network, can produce electrocardiogram (ECG) data through photoplethysmography (PPG), track present activities through a three-axis accelerometer sensor [5], and identify human falls in real-time [6].

This scoping review aims to shed light on the limitations of existing healthcare monitoring solutions and their limited adoption by older individuals and provide opportunities towards the research and development of novel unobtrusive health monitoring concepts.

2. Methods

A scoping review was conducted around the smart technologies used for health monitoring of the older population since 2010. PubMed, Scopus, WebOfScience, and Google Scholar databases were searched from 2010 to 2023. This review focused on gathering information from journal articles, papers, and reviews specifically addressing the monitoring of the older population's health through the utilization of wearable sensors and smart health devices. Search terms related to health monitoring of the older population included: "elderly", "geriatric", "older adults", "health monitoring", "behavioral monitoring", "activities of daily living", "everyday objects", "common objects", or "routine objects". The eligibility criteria for this review included articles, RCTs, observational studies, case studies and reviews that are written in English and focus on the elderly population (60 years or older). Anything not in the context of technologies used for health monitoring and anything without clear or relevant outcome measures was excluded.

3. Results

In total 282 papers were retrieved, whereas 43 ones were screened as eligible and after complete text check, 19 studies were selected to be involved in this scoping review.

3.1. Real-world applicability

Despite the remarkable potential of wearable sensors, challenges to their real-life applicability persist [7]. Issues such as limited digital literacy skills, data privacy concerns, and device performance need to be addressed for widespread adoption, particularly among older adults [8]. Passive sensors, operating without user interaction, emerge as a promising direction to reduce the burden of device interaction and enhance adherence, ensuring that health monitoring remains seamless for older adults [9].

Ros et al. (2010) presented a system that uses RFID tags as sensors to identify everyday objects in the environment. These sensors allow the system to identify every object when the person is touching them. Additionally, a hand-held device was used to collect user information and inform the right person (the user, the family, or the medical

assistance center) when something was wrong. The system aimed to monitor the Activities of Daily Living (ADLs) and provide services in case of dangerous situations while safely maintaining the independence of elders [10]. Neuhaeuser et al. (2014) introduced a portable, wearable RFID system to automatically record interactions with everyday objects, particularly aimed at aiding older individuals in independent living. The system showed promise in the automatic detection of ADLs, minimizing the need for intrusive equipment in the older person's home while preserving privacy. However, challenges remained in optimizing power consumption, readout frequency, and ensuring a reliable detection rate, particularly in real-world scenarios [11]. Civitarese et al. (2016) presented an innovative approach to unobtrusively monitor activities of daily living (ADLs) through proximity sensors attached to everyday objects. Utilizing Bluetooth Low Energy (BLE) beacons equipped with accelerometers and temperature sensors, the system was able to detect specific manipulations by seniors in their homes. Preliminary results indicated the potential utility of this method in enriching ADL recognition systems with some concerning limitations emerging, including issues with energy consumption due to high transmission rates, interference near metal objects, and vulnerability to water exposure and high temperatures, potentially limiting its real-world deployment [12]. Hu et al. (2020) presented a comprehensive approach to addressing the challenges of activity recognition and modeling in elderly care, particularly for solitary older adults. Utilizing wearable sensors such as a smartwatch along with environmentally mounted sensors like a 2D laser scanner and an RGB-D camera, they focused on developing an unobtrusive and continuous monitoring system. However, limitations of environmentally mounted sensors are acknowledged, including occlusion and coverage issues. To overcome these limitations, a sensor fusion strategy was proposed, along with an automatic labeling algorithm based on IMU signals. The study involved a cohort of healthy older adults living in a smart home environment, with activities monitored without a formal training process [13]. Balogh et al. (2022) addressed obtrusiveness by introducing an IoT device capable of categorizing emotional states non-invasively. The proposed device integrated facial recognition with physiological signal measurement to identify basic emotions, streamlining processes on microcomputers. The study highlighted the correlation between physiological signals and emotions, underscoring the device's potential for unobtrusive monitoring. Despite limitations like facial expression recognition complexity and data requirements, it offers insights for scalable and efficient emotion recognition systems while addressing privacy concerns [14].

3.2. Ambient Sensors

In the pursuit of unobtrusive health monitoring for the elderly, there is a growing recognition of the potential offered by ambient sensors. These sensors, embedded in the environment rather than worn on the body, present a less intrusive alternative to traditional wearable devices. The shift towards ambient sensors arises from the acknowledgment that, for many older adults, the adoption of wearable devices may be hindered by factors such as limited digital literacy, privacy concerns, and discomfort associated with continuous device interaction.

Suryadevara et al. (2014) introduced a pervasive computing framework aimed at assessing the well-being of independently living elderly individuals. The framework employed a wireless sensor network (WSN) and intelligent home-monitoring software to track activities of daily living (ADLs) through household object usage. The system utilized a combination of smart sensors, including passive infrared (PIR) sensors and

XBee units, strategically placed throughout the home to monitor movements and interactions with objects. These sensors enable accurate recognition of ADLs and comprehensive wellness assessment, facilitating preventive healthcare measures [15]. Bobin et al. (2018) introduced SyMPATHy, a smart cup designed to monitor stroke patients' daily activities at home. Equipped with sensors, including liquid level and movement detectors, the cup was able to track tasks like sitting, standing, walking, stairs, and drinking. Using a Support Vector Machine classifier, the system managed to achieve over 92% precision in specific activities recognition [16]. Guerrero et al. (2019) introduced a novel, IoT-based home monitoring system tailored for assisting caregivers and relatives in safeguarding elderly individuals, especially those with cognitive impairments. The system employed a network of sensors and actuators strategically placed throughout the home, including proximity sensors, RFID sensors, and Arduino boards, to detect potential risks. These sensors interact with everyday objects like lights and speakers to issue notifications to caregivers via smartphones [17]. Zolfaghari et al. (2023) proposed a novel approach to activity recognition using unobtrusive smart-home sensors and deep-learning algorithms. By mitigating the need for wearable sensors or cameras, the method aimed to preserve user privacy while still accurately detecting daily activities. The study's experimental evaluation demonstrated promising results, highlighting its potential to mitigate the obtrusiveness often associated with traditional monitoring methods. However, the paper acknowledges limitations, such as the reliance on scripted activities and sensor availability, suggesting avenues for future research to enhance applicability in real-world settings [18].

4. Discussion

The integration of wearable sensors signifies a notable advancement in health monitoring for the elderly, offering real-time tracking of vital signs and activity levels. Despite the potential benefits, challenges such as limited digital literacy and privacy concerns persist, hindering widespread adoption as literature suggests [19]. Efforts [20] to address these barriers are ongoing, with a focus on passive and ambient sensors as potential solutions. These sensors, strategically embedded within living spaces, offer a seamless and non-disruptive approach to health monitoring, aiming to enhance the well-being of older adults while preserving their independence and privacy.

5. Conclusions

The UN decade of healthy ageing highlighted the need for accessible, affordable, and long-term healthcare to the growing elderly population. This review explored some of the technologies used for health monitoring of the elderly population in the last 15 years. Current challenges towards their better adoption include obtrusiveness, limited familiarity with technology, and user acceptance. According to that, future works could focus on the human aspects of the technologies used into elderly health monitoring, as well as to explore ways to repurpose everyday objects used by the elderly in their daily activities of living, to smart health monitoring devices, so as to enhance trust and adoption in real-world settings.

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